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Environmental control system having a fuel cell assembly

Abstract

An environmental control system assembly for an aircraft is provided. The assembly includes: an environmental control system; a fuel cell assembly in electrical communication with the environmental control system for providing electrical power to the environmental control system; and a controller operably connected to the fuel cell assembly, the controller operable to modulate an amount of power generated by the fuel cell assembly and provided to the environmental control system based on load forecasting data from an ECS load forecasting module of the controller.

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Background/Summary

FIELD

(1) The present disclosure relates to an environmental control system assembly for an aircraft including an environmental control system and a fuel cell assembly.

BACKGROUND

(2) Commercial aircraft typically include an environmental control system, or ECS, to maintain a temperature and pressure within a cabin of the aircraft within a desired range. For example, it will be appreciated that modern commercial aircraft are configured to cruise at relatively high altitudes where an ambient temperature is much lower than would be reasonable for passengers within the cabin, and further where an ambient pressure is much lower than would be reasonable for passengers within the cabin. The ECS may provide an airflow to the cabin to maintain a cabin temperature and a cabin pressure within a desired range.

(3) Supplying power to the ECS can be burdensome, particularly for certain ECS locations. Accordingly, a system and method for improving the provision of power to the ECS would be welcomed in the art.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) A full and enabling disclosure of the present disclosure, including the best mode thereof, directed to one of ordinary skill in the art, is set forth in the specification, which makes reference to the appended figures, in which:

(2) FIG. 1 is a top view of an aircraft in accordance with an exemplary embodiment of the present disclosure.

(3) FIG. 2 is a schematic diagram of an environmental control system assembly in accordance with an exemplary embodiment of the present disclosure.

(4) FIG. 3 is a perspective view of a fuel cell stack of a fuel cell assembly of the exemplary environmental control system assembly of FIG. 2.

(5) FIG. 4 is a schematic view of a fuel cell of the exemplary fuel cell stack of FIG. 3.

(6) FIG. 5 is a schematic view of a controller in accordance with an exemplary embodiment of the present disclosure.

(7) FIG. 6 is a schematic diagram of an environmental control system assembly in accordance with

another exemplary embodiment of the present disclosure.

(8) FIG. 7 is a schematic diagram of an environmental control system assembly in accordance with yet another exemplary embodiment of the present disclosure.

(9) FIG. 8 is a flow diagram of a method of operating an environmental control system assembly in accordance with an exemplary aspect of the present disclosure.

(10) FIG. 9 is a schematic diagram of an environmental control system assembly in accordance with another exemplary embodiment of the present disclosure.

(11) FIG. 10 is a schematic diagram of a control scheme for environmental control system assembly in accordance with another exemplary embodiment of the present disclosure.

(12) FIG. 11 is a schematic diagram of an environmental control system assembly in accordance with an exemplary embodiment of the present disclosure.

(13) FIG. 12 is a flow diagram of a method of operating an environmental control system assembly in accordance with another exemplary aspect of the present disclosure.

DETAILED DESCRIPTION

(14) Reference will now be made in detail to present embodiments of the disclosure, one or more examples of which are illustrated in the accompanying drawings. The detailed description uses numerical and letter designations to refer to features in the drawings. Like or similar designations in the drawings and description have been used to refer to like or similar parts of the disclosure.

(15) The word “exemplary” is used herein to mean “serving as an example, instance, or illustration.” Any implementation described herein as “exemplary” is not necessarily to be construed as preferred or advantageous over other implementations. Additionally, unless specifically identified otherwise, all embodiments described herein should be considered exemplary.

(16) The singular forms “a”, “an”, and “the” include plural references unless the context clearly dictates otherwise.

(17) The terms “upstream” and “downstream” refer to the relative direction with respect to fluid flow in a fluid pathway. For example, “upstream” refers to the direction from which the fluid flows, and “downstream” refers to the direction to which the fluid flows.

(18) The present disclosure is generally related to an environmental control system assembly (“ECS assembly”) for an aircraft. The ECS assembly includes an environmental control system (“ECS”), a cabin airflow delivery system, a cabin exhaust delivery system, and a fuel cell assembly. The ECS is generally configured to generate a cabin inlet airflow and the cabin airflow delivery system is in airflow communication with the ECS for receiving the cabin inlet airflow and providing such cabin inlet airflow to a cabin of the aircraft to maintain a desired temperature and pressure within the cabin. The cabin exhaust delivery system is in airflow communication with the cabin of the aircraft for receiving a cabin exhaust airflow from the cabin.

(19) In a first exemplary embodiment, the cabin exhaust delivery system is further in airflow communication with the fuel cell assembly for providing at least a portion of the cabin exhaust airflow to the fuel cell assembly. The fuel cell assembly may be configured to generate electric power using the cabin exhaust airflow.

(20) In at least certain exemplary aspects of this first embodiment, the fuel cell assembly may be a polymer exchange membrane (PEM) fuel cell assembly including PEM fuel cells. PEM fuel cells generally require an airflow between, e.g., 60 degrees Celsius and 90 degrees Celsius. During operation of the aircraft at, e.g., a cruise operation, significant energy would be required to condition the ambient airflow to be within this temperature range. Similar constraints exist on a pressure of the airflow provided to PEM fuel cells. The inventors of the present disclosure, however, found that by utilizing the cabin exhaust airflow, little to no pre-conditioning of the airflow is required to generate electricity using the cabin exhaust airflow, providing significant efficiency benefits.

(21) In a second exemplary embodiment, the ECS assembly further includes a controller operably

connected to the fuel cell assembly. The controller may be operable to modulate an amount of power generated by the fuel cell assembly and provided to the ECS based on load forecasting data from an ECS load forecasting module of the controller. The ECS load forecasting module may determine the load forecasting data based on a passenger number for the aircraft, weather forecast data, historical data, or a combination thereof. In particular, the ECS load forecasting module may determine the load forecasting data based on the passenger number for the aircraft. The inventors of the present disclosure found a strong correlation between an amount of power required by the ECS and the passenger number. By utilizing this correlation, the load forecasting data may be determined, allowing for a controller of the fuel cell assembly to provide a desired amount of power to operate the ECS, e.g., without a need for external power.

(22) Referring now to the drawings, wherein identical numerals indicate the same elements throughout the FIGS., FIG. 1 provides a top view of an exemplary aircraft 10 as may incorporate various embodiments of the present disclosure. As shown in FIG. 1, the aircraft 10 defines a longitudinal direction L that extends therethrough, a transverse direction T, a forward end 14, and an aft end 16.

(23) Moreover, the aircraft 10 includes a fuselage 20, extending longitudinally from the forward end 14 of the aircraft 10 towards the aft end 16 of the aircraft 10, and a pair of wings 22, or rather, a first wing 22A and a second wing 22B. The first wing 22A extends outwardly from the fuselage 20 generally along the transverse direction T with respect to the longitudinal direction L, from a port side 24 of the fuselage 20. Further, the second wing 22B similarly extends outwardly from the fuselage 20, generally along the transverse direction T with respect to the longitudinal direction L, from a starboard side 26 of the fuselage 20. In addition, the aircraft 10 further includes a vertical stabilizer 32 and a pair of horizontal stabilizers 36. The fuselage 20 additionally includes an outer surface 40 and defines a cabin 42 inward of the outer surface 40 of the fuselage 20 (depicted in phantom in FIG. 1). The cabin 42 generally refers to an area to be occupied by one or more passengers, crew, or both during a flight operation of the aircraft 10. The fuselage 20, wings 22, and stabilizers 32, 36 may together be referred to as a body of the aircraft 10.

(24) However, it should be appreciated that in other exemplary embodiments of the present disclosure, the aircraft 10 may additionally or alternatively include any other suitable configuration of, e.g., stabilizers that may or may not extend directly along a vertical direction or horizontal/transverse direction T, wings 22, etc.

(25) The exemplary aircraft 10 of FIG. 1 also includes a propulsion system. The exemplary propulsion system depicted includes a plurality of aircraft engines, at least one of which is mounted to each of the pair of wings 22A, 22B. Specifically, the plurality of aircraft engines includes a first aircraft engine 44 mounted to the first wing 22A and a second aircraft engine 46 mounted to the second wing 22B. In at least certain exemplary embodiments, the aircraft engines 44, 46 may be configured as turbofan jet engines suspended beneath the wings 22A, 22B in an under-wing configuration.

(26) Alternatively, however, in other exemplary embodiments any other suitable aircraft engine may be provided. For example, in other exemplary embodiments the first and/or second aircraft engines 44, 46 may alternatively be configured as turbojet engines, turboshaft engines, turboprop engines, etc. Further, in still other exemplary embodiments, the propulsion system may additionally or alternatively include one or more electric, or hybrid-electric, aircraft engines (e.g., electric fans).

(27) The aircraft 10 may further include one or more fuel tanks 48. In at least certain exemplary aspects, as is depicted in phantom in FIG. 1, the one or more fuel tanks 48 of the aircraft 10 may be housed within the wings 22A, 22B of the aircraft 10 (depicted in phantom only in wing 22B for clarity). The one or more fuel tanks 48 may provide fuel to the aircraft engines 44, 46 of the propulsion system of the aircraft 10.

(28) Moreover, it will be appreciated that the exemplary aircraft 10 of FIG. 1 further includes an environmental control system assembly 100 (or “ECS assembly 100”) having an environmental

control system **102** (or “ECS **102**”). The ECS **102** may generally be configured to receive an airflow, such as an ambient airflow or ram airflow **104** from a ram air inlet, a bleed airflow **106** from one or both of the aircraft engines **44**, **46**, or both. The ECS **102** is further configured to condition such airflow and provide such airflow to the cabin **42** to assist with pressurizing the cabin **42** and providing thermal control of the cabin **42**, and optionally to provide cooling for accessory systems, such as avionics.

(29) Notably, for the embodiment depicted, the ECS assembly **100** is located generally at a juncture between the first wing **22A** and the fuselage **20**. However, in other exemplary embodiments, the ECS assembly **100** may additionally or alternatively be located at any other suitable location within the aircraft **10**. For example, in other exemplary embodiments, the ECS assembly **100** may be located at a juncture between the second wing **22B** and the fuselage **20**, or alternatively, as depicted in phantom, at the aft end **16** of the aircraft **10**.

(30) Briefly, it will further be appreciated that the aircraft **10** includes an aircraft controller **50**. The aircraft controller **50** may be operably coupled to one or more outside data sources for receiving data from such outside data sources relating to, e.g., a passenger number (or passenger count) for a particular flight operation, weather data for the flight operation, flight data, etc. The aircraft controller **50** may include a similar structure as the controller **250** of the ECS assembly **100** described below with reference to FIG. 5.

(31) Referring now to FIG. 2, a schematic diagram is provided of the exemplary ECS assembly **100** introduced above with reference to FIG. 1. As will be appreciated, the exemplary ECS assembly **100** is operable with the cabin **42** of the aircraft **10** to, e.g., provide cooling air to the cabin **42**.

(32) As will be appreciated, the ECS assembly **100** generally includes the ECS **102** for generating a cabin inlet airflow, a cabin airflow delivery system **108** in airflow communication with the ECS **102** for receiving the cabin inlet airflow from the ECS **102** and providing the cabin inlet airflow to the cabin **42**, and a cabin exhaust delivery system **110** in airflow communication with the cabin **42** for receiving a cabin exhaust airflow from the cabin **42**.

(33) The ECS **102** generally includes a compressed airflow source **112** and an air cycle machine **114** (enclosed by phantom lines in FIG. 2). In the embodiment shown, the compressed airflow source **112** includes an initial compressor **116** and an electric motor **118** drivingly coupled to the initial compressor **116** and the air cycle machine **114**. The initial compressor **116** is in airflow communication with an ECS inlet **120** configured to receive an ECS inlet airflow, which may be, e.g., an ambient airflow from a ram air inlet (see, e.g., ram airflow **104** of FIG. 1). The initial compressor **116** is configured to compress the ECS inlet airflow received from the ECS inlet **120**, referred to below as simply an ECS airflow. The compressed ECS airflow is then provided to the air cycle machine **114** through a first duct **122**. As will be appreciated, during certain flight operations, the ECS airflow may be relatively cool. While compressing the ECS airflow through the initial compressor **116** may increase a temperature of the ECS airflow, the ECS **102** further includes a first heat exchanger **124** in thermal communication with the first duct **122** for adding heat to the compressed ECS airflow through the first duct **122**.

(34) Briefly, it will be appreciated that the compressed airflow source **112** may be any other suitable source of compressed airflow. For example, in other embodiments, the compressed airflow source **112** may be a bleed airflow source configured to received bleed airflow from an engine of the aircraft (e.g., bleed airflow **106** depicted in FIG. 1). With such a configuration, the compressed airflow source **112** may include a pressure reducer to reduce a pressure of the bleed airflow and, optionally, a heat exchanger to modify a temperature of the bleed airflow provided to the air cycle machine **114** such that it is within an acceptable temperature range. In such a manner, the compressed airflow source **112** may generally be referred to as an airflow conditioner, as it may be configured to provide compressed airflow within the acceptable pressure (and optionally temperature) range to the air cycle machine **114**.

(35) Referring still to FIG. 2, the air cycle machine **114** includes an air cycle machine compressor

126 and an air cycle machine turbine **128**. Notably, for the embodiment shown, the initial compressor **116**, the air cycle machine compressor **126**, and the air cycle machine turbine **128** are each coupled to a common ECS shaft **130**, with the ECS shaft **130** rotatable with, and more specifically rotatably driven by, the electric motor **118**. The air cycle machine compressor **126** is in airflow communication with the first duct **122** for receiving the compressed ECS airflow, and is configured to further compress the ECS airflow and provide such ECS airflow to the air cycle machine turbine **128** through a second duct **132** of the ECS **102**.

(36) As will further be appreciated from FIG. 2, the exemplary ECS **102** depicted further includes a second heat exchanger **134** in thermal communication with the second duct **132** for adding additional heat to the ECS airflow downstream of the air cycle machine compressor **126** and upstream of the air cycle machine turbine **128**.

(37) The air cycle machine turbine **128** is in airflow communication with the second duct **132** for receiving the ECS airflow and expanding the ECS airflow, reducing a pressure and a temperature of the ECS airflow. The air cycle machine turbine **128** is in airflow communication with the cabin airflow delivery system **108** for providing the expanded ECS airflow to the cabin airflow delivery system **108** as a cabin inlet airflow.

(38) In the exemplary embodiment depicted, the ECS assembly **100**, or rather, the cabin airflow delivery system **108**, includes a mixer **136** and the ECS **102** further includes an air cycle machine turbine bypass duct **138** extending from the air cycle machine compressor **126** to the mixer **136**, bypassing the air cycle machine turbine **128** and, for the embodiment shown, the second heat exchanger **134**. The mixer **136** is configured to receive airflow from the air cycle machine turbine bypass duct **138** and incorporate such airflow into the cabin inlet airflow.

(39) A turbine bypass valve **140** is in airflow communication with the air cycle machine turbine bypass duct **138** for modulating an airflow therethrough. As will be appreciated, by modulating the amount of airflow through the air cycle machine turbine bypass duct **138**, temperature and pressure regulation of the cabin inlet airflow may be accomplished. For example, as will be appreciated, the airflow through the air cycle machine turbine bypass duct **138** may be at a higher pressure and temperature than the airflow provided from the air cycle machine turbine **128** to the cabin airflow delivery system **108**.

(40) In addition, for the embodiment shown, it will be appreciated that the ECS assembly **100** further includes a recirculation airflow path **142** and a recirculation fan **144** (or recirculation compressor). The recirculation airflow path **142** is in airflow communication with cabin **42** for receiving a portion of the cabin exhaust airflow and further is in airflow communication with the mixer **136** for providing the portion of the cabin exhaust airflow to the mixer **136**. The recirculation fan **144** is configured to increase a pressure of the portion of the cabin exhaust airflow prior to the portion of the cabin exhaust airflow reaching the mixer **136**. The mixer **136** may further be configured to incorporate the portion of the cabin exhaust airflow through the recirculation airflow path **142** back into the cabin inlet airflow prior to the cabin airflow delivery system **108** providing such cabin inlet airflow to the cabin **42**.

(41) Referring still to FIG. 2, it will be appreciated that the ECS assembly **100** further includes the cabin exhaust delivery system **110** and a fuel cell assembly **150**. The cabin exhaust delivery system **110** is in airflow communication with the cabin **42** of the aircraft **10** for receiving all or a portion of the cabin exhaust airflow and is further in airflow communication with the fuel cell assembly **150** for receiving all or the portion of the cabin exhaust airflow from the cabin exhaust delivery system **110**.

(42) In particular, it will be appreciated that the fuel cell assembly **150** generally includes a fuel delivery system **152**, an air delivery system, an electric power output **154**, and a fuel cell stack **156**.

(43) The air delivery system includes the cabin exhaust delivery system **110**, which in turn includes a fuel cell inlet line **158** and a fuel cell outlet line **160**. The fuel cell inlet line **158** is in fluid communication with the cabin **42** and the fuel cell stack **156** for receiving the cabin exhaust airflow

from the cabin **42** and providing such cabin exhaust airflow to the fuel cell stack **156**. The fuel cell outlet line **160** is in fluid communication with the fuel cell stack **156** for receiving a fuel cell exhaust airflow from the fuel cell stack **156**.

(44) Further, for the embodiment shown, the air delivery system includes a fuel cell bypass line **162** fluidly connecting the fuel cell inlet line **158** to the fuel cell outlet line **160**, bypassing the fuel cell stack **156**. A fuel cell bypass valve **164** is in airflow communication with the fuel cell bypass line **162** for modulating an amount of airflow through the fuel cell bypass line **162**. Modulation of the amount of airflow through fuel cell bypass line **162** may allow for modulation of one or more operating conditions of the fuel cell assembly **150**, as will be discussed in more detail below.

(45) Further, it will be appreciated that for the exemplary embodiment depicted, the fuel cell assembly **150** further includes a humidifier **166** in airflow communication with the fuel cell inlet line **158** and the fuel cell outlet line **160**. The humidifier **166** may be configured to extract water from the fuel cell exhaust airflow through the fuel cell outlet line **160** and provide such water to the cabin exhaust airflow through the fuel cell inlet line **158** upstream of the fuel cell stack **156**.

Depending on a chemistry of the fuel cell stack **156**, it may be desirable to have the cabin exhaust airflow provided to the fuel cell stack **156** with a threshold level of humidity.

(46) Briefly, it will be appreciated that airflow from the fuel cell outlet line **160** of the cabin exhaust delivery system **110** may further be provided through a turbine **167** connected to the common ECS shaft **130** of the ECS **102**. In such a manner, the ECS **102** may be configured to extract remaining energy within such airflow prior to dumping the airflow, e.g., to ambient.

(47) The fuel delivery system **152** of the fuel cell assembly **150** generally includes a fuel source **168**, a fuel delivery line **170**, and a fuel valve **172**. The fuel source **168** may be any aviation fuel or hydrogen. The fuel delivery line **170** extends from the fuel source **168** to the fuel cell stack **156** and the fuel valve **172** is in fluid communication with the fuel delivery line **170** for modulating an amount of fuel flow through the fuel delivery line **170**. Modulation of the fuel valve **172** may allow for modulation of one or more operating conditions of the fuel cell assembly **150**, as will also be discussed in more detail below.

(48) The fuel delivery system **152** further includes a fuel exhaust line **174** in fluid communication with fuel cell stack **156** for receiving output products of the fuel cell stack **156**.

(49) Although not depicted, the fuel cell assembly **150** may include, e.g., one or more fuel reformers in flow communication with the fuel delivery system for generating a hydrogen rich gas for the fuel cell stack **156**.

(50) The electric power output **154** is configured to receive electrical power from the fuel cell stack **156** and generally includes a power controller **176**. The power controller **176** may include, e.g., power electronics to convert or condition electrical power received from the fuel cell stack **156**. For example, the power controller **176** may include a DC/DC converter to convert the electrical power received to a desired current, voltage, or both.

(51) Moreover, referring now to FIG. **3**, a schematic illustration is provided as a perspective view of the fuel cell stack **156** of the fuel cell assembly **150** of FIG. **2**.

(52) The fuel cell stack **156** depicted includes a housing **180** having an outlet side **182** and a side **184** that is opposite to the outlet side **182**, a fuel and air inlet side **186** and a side **188** that is opposite to the fuel and air inlet side **186**, and sides **190**, **192**. The side **190**, the side **188**, and the side **184** are not visible in the perspective view of FIG. **3**.

(53) As will be appreciated, the fuel cell stack **156** may include a plurality of fuel cells **202** that are “stacked,” e.g., side-by-side from one end of the fuel cell stack **156** (e.g., fuel and air inlet side **186**) to another end of the fuel cell stack **156** (e.g., side **188**). As such, it will further be appreciated that the outlet side **182** includes a plurality of outlets **194**, each from a respective fuel cell **202** of the fuel cell stack **156**. During operation, output products **196** are directed from the outlets **194** out of the housing **180**. As will be appreciated from the description of FIG. **4**, below, the outlets **194** include separate fuel outlets (which may be in fluid communication with the fuel exhaust line **174**

(see FIG. 2)) and air outlets (which may be in fluid communication with the fuel cell outlet line **160** of the cabin exhaust delivery system **110** (see FIG. 2)).

(54) The fuel and air inlet side **186** includes one or more fuel inlets **198** and one or more air inlets **200**. Optionally, one or more of the inlets **198**, **200** can be on another side of the housing **180**. Each of the one or more fuel inlets **198** is fluidly coupled with the fuel delivery line **170** of the fuel delivery system **152**. Each of the one or more air inlets **200** is fluidly coupled with the fuel cell inlet line **158** of the air delivery system.

(55) Referring now to FIG. 4, a close-up, schematic view is provided of a fuel cell **202** of the fuel cell stack **156** of FIG. 3. As will be appreciated, fuel cells are electro-chemical devices which may convert chemical energy from a fuel into electrical energy through an electro-chemical reaction of the fuel, such as hydrogen, with an oxidizer, such as oxygen contained in the atmospheric air. Fuel cell systems may advantageously be utilized as an energy supply system because fuel cell systems may be considered environmentally superior and highly efficient when compared to at least certain existing systems. As a single fuel cell, such as the fuel cell **202** depicted in FIG. 4, may only be capable of generating on the order of one (1) volt of power, a plurality of fuel cells may be stacked together to form a fuel cell stack, such as the fuel cell stack **156** of FIG. 3, to generate a desired voltage. The exemplary fuel cell **202** depicted in FIG. 4, and each of the fuel cells **202** of the fuel cell stack **156** of FIG. 3, are configured as proton exchange membrane fuel cells (“PEM fuel cells”), also known as a polymer electrolyte membrane fuel cell. PEM fuel cells have an operating temperature range and operating temperature pressure determined to work well with the conditions described herein.

(56) More specifically as is depicted schematically in FIG. 4, the fuel cell **202** generally includes a cathode side **204**, an anode side **206**, and an electrolyte layer **208** positioned between the cathode side **204** and the anode side **206**. The cathode side **204** may generally include a cathode **210** and the anode side **206** and may generally include an anode **212**.

(57) The cathode side **204** includes a cathode inlet **214** and a cathode outlet **216** and the anode side **206** includes an anode inlet **218** and an anode outlet **220**. The cathode side **204** of the fuel cell **202**, and more specifically, the cathode inlet **214** of the cathode side **204** of the fuel cell **202**, is in fluid communication with the cabin exhaust delivery system **110**, and more specifically, the fuel cell inlet line **158** of the cabin exhaust delivery system **110** of FIG. 2. The cathode outlet **216** is in fluid communication with the fuel cell outlet line **160** of the cabin exhaust delivery system **110**.

Similarly, the anode side **206** of the fuel cell **202**, and more specifically, the anode inlet **218** of the anode side **206** of the fuel cell **202**, is in fluid communication with the fuel delivery line **170** of the fuel delivery system **152**. The anode outlet **220** is in fluid communication with the fuel exhaust line **174** of the fuel delivery system **152**.

(58) Referring now back to FIG. 2, it will be appreciated that during operation of the ECS assembly **100**, the ECS assembly **100** may operate the electric motor **118** to compress and condition the ECS airflow using the initial compressor **116** and air cycle machine **114**. The ECS assembly **100** may provide the ECS airflow as the cabin inlet airflow to and through the cabin airflow delivery system **108** to the cabin **42** of the aircraft **10**, providing for a pressurization and thermal management of the cabin **42** of the aircraft **10**.

(59) The ECS assembly **100** may then provide at least a portion of the cabin exhaust airflow to and through the cabin exhaust delivery system **110** to the fuel cell assembly **150**. The fuel cell assembly **150** may utilize the cabin exhaust airflow to generate electric power, in combination with fuel flow provided from the fuel delivery system **152**. In such a manner, the fuel cell assembly **150** may be integrated with the ECS **102** to efficiently generate electric power. More specifically, instead of the ECS **102** dumping the cabin exhaust airflow to ambient, the cabin exhaust airflow may be utilized in the useful production of electric power using the fuel cell assembly **150**.

(60) As briefly mentioned above, the fuel cell stack **156** may include a plurality of PEM fuel cells, such that the fuel cell assembly **150** may be configured as a PEM fuel cell assembly. The cabin

exhaust airflow may be at a temperature and pressure particularly desirable for generating electric power using PEM fuel cells and a PEM fuel cell assembly, such as the fuel cell assembly **150** depicted in FIG. **2** (described also with respect FIGS. **3** and **4**). In particular, the cabin exhaust airflow may be at a temperature between 60 degrees C. and 90 degrees C., such as between 70° C. and 80° C. Further, a pressure of the cabin exhaust airflow may be between 60 kilopascals and 100 kilopascals such as between 70 kilopascals and 90 kilopascals. These conditions may be desirable for generating electric power using a PEM fuel cell. Accordingly, by utilizing such airflow, the fuel cell assembly **150** need not use excess power to specifically condition an airflow for the fuel cell assembly **150**.

(61) Further, referring still to FIG. **2**, it will be appreciated that the fuel cell assembly **150** may be configured to generate electric power to sustain operations of the ECS **102** of the ECS assembly **100**, without requiring an electric power source outside of the ECS assembly **100**.

(62) In particular, for the embodiment depicted, the electric motor **118**, which as noted above is configured to drive operations of the ECS assembly **100**, defines a maximum power draw. As used herein, the term “maximum power draw” refers to the maximum amount of electric power required for a particular component during all anticipated non-failure mode and non-emergency mode operations for the particular component.

(63) Further, for the embodiment depicted, the fuel cell assembly **150** defines a maximum power rating. As used herein, the term “maximum power rating” refers to a maximum amount of electric power that may be generated during anticipated operating conditions without pre-maturely wearing or degrading the component. For example, the maximum power rating may refer to the maximum amount of power that the fuel cell assembly **150** may generate during a cruise operation of the aircraft **10** within which it is incorporated.

(64) For the exemplary embodiment depicted, the maximum power rating of the fuel cell assembly **150** is higher than the maximum power draw of the electric motor **118**. In such a manner, all of the electric power required for operating the ECS **102** may be provided to the ECS **102** from the fuel cell assembly **150**. Such may allow for the ECS assembly **100** to be located at a position within the aircraft **10** remote from one or more power sources without requiring relatively heavy, expensive, and complicated electric communication buses.

(65) It will further be appreciated that in at least certain exemplary aspects, depending on how much higher the maximum power rating of the fuel cell assembly **150** is than the maximum power draw, the fuel cell assembly **150** may further be in communication with an electric power bus **222** of the aircraft **10** for providing excess electric power to the electric power bus **222**.

(66) Referring still to FIG. **2**, it will be appreciated that the ECS assembly **100** further includes a controller **250**. The controller **250** is in operable communication with various aspects of the ECS assembly **100** for controlling certain operations of the ECS assembly **100**. For example, the ECS assembly **100** may include one or more sensors for sensing various operating conditions. For example, in the embodiment depicted the ECS assembly **100** includes a fuel cell sensor **253** configured to sense data indicative of various operating parameters of the fuel cell assembly **150**, such as of the fuel cell stack **156**. For example, the fuel cell sensor **253** may be configured to sense data indicative of a temperature of the fuel cell stack **156**, a pressure of one or more flows to, through, from, or around the fuel cell stack **156**, a gas composition of various flows to, through, from, or around the fuel cell stack **156**, etc. In addition, the ECS assembly **100** includes a cabin exhaust airflow sensor **254** configured to sense data indicative of the cabin exhaust airflow through the cabin exhaust delivery system **110**, such as, e.g., a flow rate of the cabin exhaust airflow, a pressure of the cabin exhaust airflow, a temperature of the cabin exhaust airflow, a gas composition of the cabin exhaust airflow, etc. Further, still, the ECS assembly **100** includes a cabin sensor **256** for sensing data indicative of an environment within the cabin **42**, such as, e.g., a temperature within the cabin **42**, a gas composition within the cabin **42**, a pressure within the cabin **42**, etc.

(67) In the embodiment depicted, the controller **250** is operably coupled to these sensors **252**, **254**,

256 and various other components of the ECS assembly **100** to control various aspects of the ECS assembly **100**. For example, in the embodiment shown, the controller **250** is further in operable communication with the power controller **176** of the fuel cell assembly **150**, the electric motor **118** of the ECS **102**, the turbine bypass valve **140**, the fuel cell bypass valve **164**, and the fuel valve **172** of the fuel cell assembly **150**.

(68) Referring now to FIG. 5, a schematic view is provided of the exemplary controller **250** of the ECS assembly **100** of FIG. 2. As noted, the exemplary controller **250** is configured to receive the data sensed from the one or more sensors (e.g., sensors **252**, **254**, **256**) and, e.g., may make control decisions for the ECS assembly **100** based on the data received from the one or more sensors.

(69) In one or more exemplary embodiments, the controller **250** depicted in FIG. 5 may be a stand-alone controller for the ECS assembly **100**, or alternatively, may be integrated into one or more other controllers, such as a controller for an aircraft **10** with which the ECS assembly **100** is integrated (e.g., aircraft controller **50** of FIG. 1), etc.

(70) Referring particularly to the operation of the controller **250**, in at least certain embodiments, the controller **250** can include one or more computing device(s) **252**. The computing device(s) **252** can include one or more processor(s) **252A** and one or more memory device(s) **252B**. The one or more processor(s) **252A** can include any suitable processing device, such as a microprocessor, microcontroller, integrated circuit, logic device, and/or other suitable processing device. The one or more memory device(s) **252B** can include one or more computer-readable media, including, but not limited to, non-transitory computer-readable media, RAM, ROM, hard drives, flash drives, and/or other memory devices.

(71) The one or more memory device(s) **252B** can store information accessible by the one or more processor(s) **252A**, including computer-readable instructions **252C** that can be executed by the one or more processor(s) **252A**. The instructions **252C** can be any set of instructions that when executed by the one or more processor(s) **252A**, cause the one or more processor(s) **252A** to perform operations. In some embodiments, the instructions **252C** can be executed by the one or more processor(s) **252A** to cause the one or more processor(s) **252A** to perform operations, such as any of the operations and functions for which the controller **250** and/or the computing device(s) **252** are configured, the operations for operating an ECS assembly **100** and/or fuel cell assembly **150** (e.g., method **280**, method **400**), as described herein, and/or any other operations or functions of the one or more computing device(s) **252**. The instructions **252C** can be software written in any suitable programming language or can be implemented in hardware. Additionally, and/or alternatively, the instructions **252C** can be executed in logically and/or virtually separate threads on the one or more processor(s) **252A**. The one or more memory device(s) **252B** can further store data **252D** that can be accessed by the one or more processor(s) **252A**. For example, the data **252D** can include data indicative of power flows, data indicative of engine/aircraft **10** operating conditions, and/or any other data and/or information described herein.

(72) The computing device(s) **252** can also include a network interface **252E** used to communicate, for example, with the other components of the ECS assembly **100**, the aircraft **10** incorporating the ECS assembly **100**, etc. For example, in the embodiment depicted, as noted above, the ECS assembly **100** includes one or more sensors for sensing data indicative of one or more parameters of the ECS assembly **100**. The controller **250** of the ECS assembly **100** is operably coupled to the one or more sensors through, e.g., the network interface **252E**, such that the controller **250** may receive data indicative of various operating parameters sensed by the one or more sensors during operation. Further, for the embodiment shown the controller **250** is operably coupled various aspects of the ECS assembly **100**, as noted above, e.g., through the network interface **252E**. In such a manner, the controller **250** may be configured control various aspects of the ECS assembly **100** in response to, e.g., the data sensed by the one or more sensors.

(73) The network interface **252E** can include any suitable components for interfacing with one or more network(s), including for example, transmitters, receivers, ports, controller **250**, antennas,

and/or other suitable components.

(74) The technology discussed herein makes reference to computer-based systems and actions taken by and information sent to and from computer-based systems. One of ordinary skill in the art will recognize that the inherent flexibility of computer-based systems allows for a great variety of possible configurations, combinations, and divisions of tasks and functionality between and among components. For instance, processes discussed herein can be implemented using a single computing device or multiple computing devices working in combination. Databases, memory, instructions, and applications can be implemented on a single system or distributed across multiple systems. Distributed components can operate sequentially or in parallel.

(75) It will be appreciated, however, that the exemplary ECS assembly **100** depicted in FIG. 2 is provided by way of example only. In other exemplary embodiments, the ECS assembly **100**, and various aspects of ECS assembly **100**, may be configured in any other suitable manner. For example, referring now to FIG. 6, a schematic, flow diagram is provided of an ECS assembly **100** in accordance with another exemplary aspect of the present disclosure. The exemplary ECS assembly **100** of FIG. 6 may be configured in substantially the same manner as exemplary ECS assembly **100** of FIG. 2. Accordingly, the same or similar numbers may refer to the same or similar parts.

(76) As depicted, the ECS assembly **100** generally includes an ECS **102** and a fuel cell assembly **150**. The ECS **102** generally includes a compressed airflow source **112** and an air cycle machine **114**. The compressed airflow source **112** is configured as an initial compressor **116** and an electric motor **118** for driving the initial compressor **116**. The air cycle machine **114** includes an air cycle machine compressor **126**. In the embodiment shown, the fuel cell assembly **150** is configured to receive a cabin exhaust air flow through a cabin exhaust delivery system **110** from a cabin **42** of an aircraft **10** (see FIG. 1).

(77) However, for the embodiment depicted, the fuel cell assembly **150** is further in airflow communication with a compressor of the ECS **102**, and more specifically, is in airflow communication with the initial compressor **116** and the air cycle machine compressor **126** of the ECS **102**. For example, the ECS assembly **100** depicted includes an initial compressor bleed duct **258** and an air cycle machine compressor bleed duct **260**, each in flow communication with a mixer valve **262** of the cabin exhaust delivery system **110**. Notably, for the embodiment shown, the ECS assembly **100** further includes a first bleed duct valve **264** in fluid communication with the initial compressor bleed duct **258** and a second bleed duct valve **266** in fluid communication with the air cycle machine bleed duct **260**. The first bleed duct valve **264** and the second bleed duct valve **266** are each in operable communication with a controller **250** of the ECS assembly **100**.

(78) Such a configuration may allow for, e.g., greater power generation during certain operating conditions, temperature and flow control of an airflow to the fuel cell assembly **150**, etc. Moreover, inclusion of one or both of the initial compressor bleed duct **258** (and the first bleed duct valve **264**) and an air cycle machine compressor bleed duct **260** (and the second bleed duct valve **266**) may avoid or alleviate a surge through the initial compressor **116** and/or air cycle machine **114** in the event other anti-surge means (such as a variable diffuser that controls the airflow which is compressed mechanically at the air intake to pass through the compressor, an Add Heat Valve (AHV) that controls the temperature of the airflow to pass through the compressor, etc.) have reached, or are approaching, their performance limits.

(79) Further, in other exemplary embodiments, still other configurations may be provided for the ECS assembly **100**. For example, referring now to FIG. 7, a schematic, flow diagram is provided of an ECS assembly **100** in accordance with yet another exemplary aspect of the present disclosure. The exemplary ECS assembly **100** of FIG. 7 may be configured in substantially the same manner as exemplary ECS assemblies **100** of FIG. 2 or 6. Accordingly, the same or similar numbers may refer to the same or similar parts.

(80) As depicted, the ECS assembly **100** generally includes an ECS **102** and a fuel cell assembly

150. The ECS **102** generally includes a compressed airflow source **112** and an air cycle machine **114**. The compressed airflow source **112** is configured as an initial compressor **116** and electric motor **118**. The air cycle machine **114** includes an air cycle machine compressor **126**. In the embodiment shown, a cabin airflow delivery system **108** is provided in airflow communication with the ECS **102** and a cabin **42** for providing a cabin inlet airflow from the ECS **102** to the cabin **42**. Further, a cabin exhaust delivery system **110** is provided in airflow communication with the cabin **42** and the fuel cell assembly **150** for providing a cabin exhaust airflow from the cabin **42** to the fuel cell assembly **150**. The fuel cell assembly **150** is configured to receive a cabin exhaust airflow through the cabin exhaust delivery system **110** from the cabin **42** of an aircraft **10** (see FIG. 1).

(81) However, for the exemplary embodiment of FIG. 7, the cabin exhaust delivery system **110** includes a recirculation fan **144** in airflow communication with the cabin exhaust delivery system **110** and the recirculation flowpath **142**. The recirculation fan **144** is configured for increasing a pressure of an air flow therethrough at a location upstream of the fuel cell assembly **150**, and the recirculation flowpath **142** extends from the cabin exhaust delivery system **110** from a location downstream of the recirculation fan **144**. The recirculation airflow path **142** extends from the cabin exhaust delivery system **110** to the mixer **136** of the cabin airflow delivery system **108**.

(82) Notably, with such exemplary embodiment, the ECS assembly **100** may further include an exhaust from the cabin **42** to ambient (not labeled). Further, although not depicted in the exemplary embodiment of FIG. 7, the ECS assembly **100** may additionally include a second cabin exhaust delivery system (independent of the cabin exhaust delivery system **110**) with a separate fuel cell assembly in airflow communication therewith (e.g., a combination of the embodiments of FIG. 2 or 6 with the embodiment of FIG. 7).

(83) Referring now to FIG. 8, a flow diagram of a method **280** of operating a fuel cell assembly is provided. The method **280** may be utilized with one or more of the exemplary fuel cell assemblies **150** (and ECS assemblies **100**) described hereinabove.

(84) The method includes at (282) providing a cabin exhaust airflow from a cabin of an aircraft through a cabin exhaust delivery system to a fuel cell assembly.

(85) The method **282** further includes at (284) generating electrical power with the fuel cell assembly using the cabin exhaust airflow.

(86) In at least certain exemplary aspects, providing the cabin exhaust airflow from the cabin of the aircraft through the cabin exhaust delivery system to the fuel cell assembly at (282) includes at (286) providing the cabin exhaust delivery system to the fuel cell assembly at a temperature of between 60 degrees Celsius and 90 degrees Celsius, and at a pressure of between 60 kilopascals and 100 kilopascals.

(87) The method **282** may further incorporate various other aspects of an ECS assembly/ies **100** described herein and may perform various other operations for which the ECS assembly/ies **100** described herein are configured to perform.

(88) Referring now to FIG. 9, an ECS assembly **100** for an aircraft **10** in accordance with another exemplary aspect of the present disclosure is provided. The exemplary ECS assembly **100** of FIG. 9 may be configured in a similar manner as exemplary ECS assembly **100** described above with reference to FIG. 2, and may be incorporated into an aircraft **10** configured in a similar manner as exemplary aircraft **10** of FIG. 1.

(89) For example, the exemplary ECS assembly **100** generally includes an ECS **102**. The ECS **102** generally includes a compressed airflow source **112** and an air cycle machine **114**. In the embodiment shown, the compressed airflow source **112** includes an initial compressor **116** and an electric motor **118** drivingly coupled to the initial compressor **116**. The initial compressor **116** is configured to receive an ECS inlet airflow, which may be a ram airflow (see, e.g., ram airflow **104** of FIG. 1), and compress the ECS inlet airflow. The compressed ECS inlet airflow is then provided to the air cycle machine **114** as an ECS airflow. The air cycle machine **114**, depicted schematically,

may include an air cycle machine compressor and an air cycle machine turbine (not shown, see, e.g., air cycle machine compressor **126** and air cycle machine turbine **128** of FIG. 2). The initial compressor **116**, the air cycle machine compressor, and the air cycle machine turbine may each coupled to a common ECS shaft **130** (see, e.g., FIG. 2), with the ECS shaft **130** rotatable with, or rather rotatable by, the electric motor **118**. The initial compressor **116** and the air cycle machine compressor may be referred to collectively as a compressor section of the ECS **102**.

(90) Notably, as discussed above with reference to FIG. 2, the compressed airflow source **112** may alternatively be configured to receive a bleed airflow from an engine. In such a configuration, the compressed airflow source **112** (also referred to herein as an airflow conditioner) may include a pressure reducer (e.g., an expansion turbine) to reduce a pressure of the bleed airflow received from the engine.

(91) Moreover, for the embodiment depicted, the ECS assembly **100** includes a cabin airflow delivery system **108** in airflow communication with the ECS **102** for receiving a cabin inlet airflow from the ECS **102** and providing such cabin inlet airflow to the cabin **42** of the aircraft **10** (see FIG. 1). At least a portion of a cabin exhaust airflow from the cabin **42** of the aircraft **10** may be recycled through a recirculation airflow path **142** and provided back to a mixer in airflow communication with the cabin airflow delivery system **108**. For the embodiment shown, a recirculation fan **144** is positioned in airflow communication with the recirculation airflow path **142** for increasing a pressure of the airflow through the recirculation airflow path **142**.

(92) Further, for the embodiment depicted, the ECS assembly **100** additionally includes a fuel cell assembly **150**. The fuel cell assembly **150** is in electrical communication with the ECS **102** for providing electrical power to the ECS **102**. More specifically, for the embodiment depicted, the fuel cell assembly **150** includes a fuel cell stack **156** and an electric power output **154** in electrical communication with the fuel cell stack **156** and further in electrical communication with the electric motor **118** of the ECS **102**. In the embodiment depicted, the fuel cell assembly **150** includes a power controller **176**. The power controller **176** may include, e.g., power electronics to convert or condition electrical power received from the fuel cell stack **156**. For example, the power controller **176** may include a DC/DC converter to convert the electrical power received to a desired current, voltage, or both.

(93) Further, for the embodiment depicted, the fuel cell assembly **150** includes a fuel source **168** configured to provide a fuel flow to the fuel cell stack **156**, and further includes a fuel exhaust line **174**.

(94) Moreover, for the embodiment depicted, the fuel cell stack **156** is configured to receive an airflow from the ECS **102**. More specifically, the ECS assembly **100** further includes a cabin exhaust delivery system **110** in airflow communication with the cabin **42** of the aircraft **10** for receiving a cabin exhaust airflow from the cabin **42** of the aircraft **10**. In the embodiment depicted, the fuel cell assembly **150**, and more specifically, the fuel cell stack **156** of the fuel cell assembly **150**, is in airflow communication with the cabin exhaust delivery system **110** for receiving at least a portion of the cabin exhaust airflow from the cabin **42** of the aircraft **10**. In the embodiment depicted, the ECS assembly **100** includes a fuel cell bypass line **162** in airflow communication with the cabin exhaust delivery system **110** allowing at least a portion of the cabin exhaust airflow to be bypassed around the fuel cell stack **156** of the fuel cell assembly **150**. The fuel cell assembly **150** further includes a fuel cell bypass valve **164** in airflow communication with the fuel cell bypass line **162** to, e.g., facilitate a control of an airflow to the fuel cell stack **156** of the fuel cell assembly **150**.

(95) It will be appreciated, however, that in other exemplary embodiments of the present disclosure, the fuel cell assembly **150** may be in airflow communication with the ECS **102** in any other suitable manner. For example, as is depicted in phantom, in other exemplary embodiments, a fuel cell stack of a fuel cell assembly **150-1** may be in airflow communication with the recirculation airflow path **142** for receiving a recirculation airflow from the cabin **42**, upstream of

the mixer **136**. Additionally, or alternatively, in other exemplary embodiments, as is also depicted in phantom, a fuel cell stack of a fuel cell assembly **150-2** may be in airflow communication with the compressor section of the ECS **102** for receiving a pressurized airflow from the ECS **102**. Further, in still other exemplary embodiments, a fuel cell stack of a fuel cell assembly **150-3** may not be in airflow communication with the ECS **102**, and as is again depicted in phantom, the fuel cell assembly **150** may include a fuel cell stack configured to receive an airflow from an airflow source **268** separate from the ECS **102**.

(96) Referring still to FIG. **9**, it will be appreciated that the ECS assembly **100** further includes a controller **250** operably connected to the fuel cell assembly **150**. More specifically, the controller **250** may be operable to modify an operating condition of the fuel cell assembly **150**. In particular, the controller **250** may be operably connected to the fuel source **168** of the fuel cell assembly **150** for controlling a fuel flow to the fuel cell stack **156**, may be operably connected to an air source of the fuel cell assembly **150** for controlling an airflow to the fuel cell stack **156** (e.g., in the embodiment of FIG. **9**, the controller **250** is operably connected to the fuel cell bypass valve **164**), may be operably connected to the electric power output **154** (e.g., the power controller **176**) of the fuel cell assembly **150** for controlling a power output (e.g., a current output) of the fuel cell stack **156**, etc.

(97) As will be appreciated, in such a manner, the controller **250** may be operable to modify a fuel cell temperature of the fuel cell assembly **150**, a current output of the fuel cell assembly **150**, a fuel utilization of the fuel cell assembly **150**, an air utilization of the fuel cell assembly **150**, or any combination thereof.

(98) As is also depicted in FIG. **9**, it will be appreciated that the controller **250** may be configured to receive data from a source **270** outside of the ECS assembly **100**. The source **270** may be, e.g., an aircraft controller **50** of the aircraft **10** within which the ECS assembly **100** is incorporated (see FIG. **1**), a ground data source, etc.

(99) In particular, for the embodiment depicted, the controller **250** is operably connected to the fuel cell assembly **150**, and more specifically, is operable to modulate an amount of power generated by the fuel cell assembly **150** and provided to the ECS **102** (from the fuel cell assembly **150**) based on load forecasting data from an ECS load forecasting module of the controller **250**. The load forecasting data may be indicative of an anticipated electrical load that will be required by the ECS **102**. In such a manner, the controller **250** may allow for the fuel cell assembly **150** to generate and provide an amount of power that will be required by the ECS **102** for an amount of time in the future. Such may reduce or eliminate a lag between a power being generated by the fuel cell assembly **150** and a power being needed by the ECS **102** to operate in a desired manner.

(100) In particular, referring now to FIG. **10**, a schematic view is provided of an ECS assembly **100** in accordance with an exemplary aspect of the present disclosure. The ECS assembly **100** of FIG. **10** may be configured in a similar manner as the exemplary ECS assembly **100** described above with reference FIG. **9**. For example, as is depicted, the ECS assembly **100** generally includes an ECS **102**, a fuel cell assembly **150** in electrical communication with the ECS **102** for providing electrical power **300** to the ECS **102**, and a controller **250** operably connected to the fuel cell assembly **150** for providing an operating condition command **302** to the fuel cell assembly **150**. The controller **250** is generally configured to receive one or more load forecasting factors **304**. Moreover, for the embodiment depicted, the controller **250** may be configured to receive a command **306**, such as a command from an aircraft controller **50** (see FIG. **1**) to operate in a prescribed manner in response to an operating condition of the aircraft **10** (e.g., taxi, takeoff, climb, cruise, descent, etc.). For example, the command **306** may be a command to operate the fuel cell assembly **150** at a power level commensurate with an operating condition of the aircraft **10**.

(101) The controller **250** is configured to receive the one or more load forecasting factors **304** and determine load forecasting data. Based on the load forecasting data, the controller **250** is configured to send a signal to the fuel cell assembly **150** to control an amount of power generated by the fuel

cell assembly **150** (the operating condition command **302**). The fuel cell assembly **150** is configured to operate according to the operating condition command **302** and provide the electrical power **300** to the ECS **102**. The ECS assembly **100** is further configured to provide a feedback signal **310** back to the controller **250** indicative of an operating condition of the ECS **102**. The feedback signal **310** may be received by the controller **250** and the controller **250** may modify the operating condition command **302** sent to the fuel cell assembly **150** in response to the feedback signal **310** indicative of actual operating conditions of the ECS **102**. In particular, the ECS assembly **100** includes an ECS sensor **312** configured to sense data provided to the controller **250** as the feedback signal **310**.

(102) A feedback signal **316** is provided to the controller **250** indicative of the one or more operating conditions of the fuel cell assembly **150**. The feedback signal **316** may be received by the controller **250** and the controller **250** may modify the operating condition command **302** sent to the fuel cell assembly **150** in response to the feedback signal **316** indicative of actual operating conditions of the fuel cell assembly **150**. In particular, the fuel cell assembly **150** further includes a fuel cell assembly sensor **314** configured to sense data indicative of one or more operating conditions of the fuel cell assembly **150**, the data provided to the controller **250** as the feedback signal **316**.

(103) The data sensed by the ECS sensor **312** indicative of the operating conditions of the ECS **102** may include data indicative of, e.g., a rotational speed of one or more components of the ECS **102**, a temperature of an airflow provided to a cabin **42** of the aircraft **10** (see FIG. **1**), a flow rate of the air flow provided to the cabin **42** of the aircraft **10** (see FIG. **1**), etc.

(104) The data sensed by the fuel cell sensor **314** indicative of the operating conditions of the fuel cell assembly **150** may include data indicative of, e.g., a temperature, humidity, flowrate, pressure, electrical current, electrical voltage of the fuel cell assembly **150**.

(105) In such a manner, it will be appreciated that the controller **250** may be operable to modulate the amount of power generated by the fuel cell assembly **150** and provided to the ECS **102** based on a feedback control scheme, a feedforward control scheme, or both. As used herein, the term “feedback control scheme” refers to a system which tends to maintain a prescribed relationship of one system variable to another by comparing functions of these variables and using the difference as a means of control. Further, as used herein, the term “feedforward control scheme” refers to a system that passes a controlling signal from a source in its external environment to a load elsewhere in its external environment.

(106) More specifically, for the embodiment shown, the controller **250** is operable to modulate the amount of power generated by the fuel cell assembly **150** and provided to the ECS **102** based on a feedback control scheme.

(107) More particularly, referring now to FIG. **11**, a schematic view is depicted of a controller **250** of an ECS assembly **100** in accordance with an exemplary aspect of the present disclosure is provided. The controller **250** and ECS assembly **100** may be configured in a similar manner as one or more of the controllers **250** and ECS assemblies **100** described above with reference to, e.g., FIGS. **9** and **10**.

(108) In particular, it will be appreciated that the controller **250** generally includes an ECS load forecasting module **320**. Further, the controller **250** is configured to receive data from one or more external sources and is configured to provide such data to the ECS load forecasting module **320** to generate load forecasting data **322** of the ECS **102**. The load forecasting data **322** of the ECS **102** may include data indicative of an anticipated electrical load of the ECS **102**, as described further below. For example, in the embodiment depicted, the ECS load forecasting module **320** is configured to receive one or more load forecasting factors **304**.

(109) The ECS load forecasting module **320** may utilize the load forecasting factors **304** to determine the load forecasting data **322**. In particular, in the embodiment depicted the ECS load forecasting module **320** includes a load forecasting model **324**, which may include, e.g., one or

more data driven models (such as neural network, fuzzy logic, a lookup table, or any combination thereof), or the like. The ECS load forecasting module **320** may receive the load forecasting factors **304**, and may utilize the load forecasting model **324** with the load forecasting factors **304** to determine the load forecasting data **322** of the ECS **102**.

(110) The one or more load forecasting factors **304** may be, e.g., a passenger number for an aircraft **10** incorporating ECS assembly **100** having the controller **250** (see, e.g., FIG. **1**). The passenger number refers to a number of passengers determined to be (or estimated to be) on the aircraft **10** (see FIG. **1**) for the current or subsequent flight operation. In such a manner, the ECS load forecasting module **320** may utilize the passenger number for the aircraft **10** (see FIG. **1**) to determine the load forecasting data **322** of the ECS **102**. As will be appreciated from the description hereinabove, the ECS **102** is operable to provide airflow to the cabin **42** of the aircraft **10** (see FIG. **1**) to maintain the cabin **42** of the aircraft **10** at a desired pressure and temperature during a flight operation of the aircraft **10**. A primary driver for an amount of airflow needed, and an amount of thermal control for the cabin **42** needed, during a particular flight operation is the number of passengers the aircraft **10** is carrying during the particular flight operation. Accordingly, by receiving this information, the ECS load forecasting module **320**, and more specifically, the load forecasting model **324** of the ECS load forecasting module **320**, may be operable to relatively accurately determine operating characteristics for the ECS **102** that will be needed during the flight operation, and therefore an amount of electrical power that will be needed to support the ECS **102** operating according to the determined operating characteristics during the flight operation.

(111) Notably, the load forecasting factors **304** may further include weather forecast data for, e.g., the flight operations of the aircraft **10** (see FIG. **1**). For example, if ambient temperatures are higher or lower than a baseline temperature, such may affect the amount of thermal control required for the cabin **42** of the aircraft **10** during the flight operations. Further, the load forecasting factors **304** may further include historical data, such as historical data for the flight operations of the aircraft **10**. The historical data may further influence the load forecasting data **322** generated by the ECS load forecasting module **320**.

(112) The ECS load forecasting module **320** is configured to provide the load forecasting data **322** to an operating condition generator module **326** of the controller **250**. The operating condition generator module **326** is configured to determine an operating condition command **302** (labeled “FC Operating Condition” in FIG. **11**) for the fuel cell assembly **150** in response to, or based off of, the load forecasting data **322** determined by the ECS load forecasting module **320**. The operating condition command **302** is provided to the fuel cell assembly **150** from the controller **250** such that the fuel cell assembly **150** may operate in a desired manner to generate and provide the electrical power **300** to the ECS **102** needed to sustain operation of the ECS **102** as anticipated by the ECS load forecasting module **320**.

(113) Notably, for the embodiment depicted, the controller **250** further includes an objective function module **328**. The objective function module **328** may be operable between one or more of the following objective functions **330**: maximum efficiency, maximum power output, maximum component life, and minimum emissions output. The objective function module **328** is in operable communication with the operating condition generator module **326**. The objective function module **328**, based on a selected objective function **330**, may influence a value of the operating condition command **302** generated by the operating condition generator module **326** to achieve the selected objective function **330**.

(114) The controller **250** is further configured to receive data indicative of a command **306**. As noted with the embodiment of FIG. **10**, the command **306** may be indicative of a baseline power output for the fuel cell assembly **150** based on, e.g., an operating condition of the aircraft **10** (see FIG. **1**). In such a case, data from the command **306** may be received by the operating condition generator module **326**. Additionally, or alternatively, the command **306** may be a command to select an objective function **330** of the objective function module **328**. In such a scenario, data from

the command **306** may be received by the objective function module **328**.

(115) Further, for the embodiment depicted, the controller **250** is configured to receive feedback data **332**. The feedback data **332** may be feedback data of the ECS **102**, such as an operating condition of the ECS **102** (see, e.g., feedback signal **310** and ECS sensor **312** of FIG. **10**). More specifically, for the embodiment depicted, the operating condition generator module **326** is configured to receive the feedback data **332**. The feedback data **332** may influence a value of the operating condition command **302** generated by the operating condition generator module **326**.

(116) Utilizing the load forecasting factors **304** and, e.g., the load forecasting model **324**, the ECS load forecasting module **320** may define an anticipated forecast timing of at least two minutes ahead and up to five hours ahead. For example, the anticipated forecast timing may be at least five minutes ahead and up to one hour ahead. In such a manner, the controller **250** may be operable to ensure the fuel cell assembly **150** is operating in a desired manner to generate and provide the electrical power **300** to the ECS **102** needed to sustain operation of the ECS **102** as anticipated by the ECS load forecasting module **320**.

(117) Accordingly, it will be appreciated that for the exemplary controller **250** depicted in FIG. **11**, to modulate the amount of power generated by the fuel cell assembly **150** and provided to the ECS **102**, the controller **250** is operable to modify an operating condition of the fuel cell assembly **150** by providing the fuel cell assembly **150** the operating condition command **302**. In such a manner, the ECS assembly **100** may be capable of providing a desired amount of electrical power to the ECS **102** to allow the ECS **102** to operate at a desired output level despite a lag in responsiveness of the fuel cell assembly **150** as compared to changes in a power demand of the ECS **102** during certain operating conditions and across certain operating conditions. Such may allow for the ECS assembly **100** to be powered solely by the fuel cell assembly **150**, without requiring additional power source(s) in electrical communication with the ECS **102** during operation of the aircraft **10**.

(118) Referring now briefly to FIG. **12**, a method **400** of operating an ECS assembly for an aircraft is provided. The method **400** may be utilized with one or more of the exemplary ECS assemblies **100** described above.

(119) The method **400** includes **(402)** determining ECS load forecasting data for an ECS of the ECS assembly. Determining the ECS load forecasting data for the ECS at **(402)** includes at **(404)** receiving data from a source external of the ECS assembly. The data received at **(404)** may include a passenger number for a flight operation (i.e., a number of passengers within the aircraft during the flight operation). In at least certain exemplary aspects, determining the ECS load forecasting data for the ECS at **(402)** includes at **(406)** determining the ECS load forecasting data utilizing an ECS load forecasting module of a controller.

(120) The method **400** further includes at **(408)** controlling a fuel cell assembly in electrical communication with the ECS to modulate an amount of power generated by the fuel cell assembly and provided to the ECS in response to the determined ECS load forecasting data at **(402)**.

(121) Further, for the exemplary aspect of the method **400** depicted, the method **400** further includes at **(410)** providing all of an electric power consumed by the ECS during a flight operation from the fuel cell assembly. The flight operation may include an entire flight envelope of a flight of an aircraft. Alternatively, the flight operation may include a segment of the flight envelope, such as a takeoff portion of the flight envelope, a climb portion of the flight envelope, a cruise portion of the flight envelope, a descent portion of the flight envelope, etc.

(122) The method **400** may further incorporate various other aspects of an ECS assembly/ies **100** described herein and may perform various other operations for which the ECS assembly/ies **100** described herein are configured to perform.

(123) Further aspects are provided by the subject matter of the following clauses:

(124) An environmental control system assembly for an aircraft, the assembly comprising: an environmental control system; a cabin airflow delivery system in airflow communication with the environmental control system for receiving a cabin inlet airflow from the environmental control

system; a cabin exhaust delivery system configured to be in airflow communication with a cabin of the aircraft for receiving a cabin exhaust airflow when the environmental control system assembly is installed in the aircraft; and a fuel cell assembly in airflow communication with the cabin exhaust delivery system for receiving the cabin exhaust airflow.

(125) The assembly of any preceding clause, wherein the low temperature fuel cell assembly is a polymer exchange membrane fuel cell assembly.

(126) The assembly of any preceding clause, wherein the fuel cell assembly comprises a fuel cell having an anode and a cathode, and wherein the cathode of the fuel cell is in airflow communication with the cabin exhaust delivery system for receiving the cabin exhaust airflow.

(127) The assembly of any preceding clause, wherein the environmental control system comprises an airflow conditioner and an air cycle machine, and wherein the environmental control system assembly further comprises a mixer.

(128) The assembly of any preceding clause, wherein the airflow conditioner is configured to receive an ambient airflow, a bleed airflow from an engine, or both.

(129) The assembly of any preceding clause, wherein the environmental control system further comprises an electric motor drivingly coupled to the compressor and the air cycle machine, wherein the electric motor defines a maximum power draw, wherein the fuel cell assembly defines a maximum power rating, and wherein the maximum power rating is higher than the maximum power draw.

(130) The assembly of any preceding clause, wherein the fuel cell assembly is in electrical communication with the electric motor of the environmental control system for providing electrical power to the electric motor of the environmental control system.

(131) The assembly of any preceding clause, wherein the environmental control system comprises a compressor and an air cycle machine, and wherein the environmental control system assembly further comprises a mixer, wherein the mixer is in airflow communication with cabin airflow delivery system, wherein the environmental control system assembly further comprises a recirculation airflow path extending from the cabin to the mixer.

(132) The assembly of any preceding clause, further comprising: a recirculation fan in airflow communication with the cabin exhaust delivery system, the recirculation airflow path, or both for increasing a pressure of an airflow therethrough.

(133) An aircraft comprising: a body comprising a fuselage and one or more wings, the fuselage defining a cabin; and an environmental control system assembly positioned within the body of the aircraft, the environmental control system assembly comprising: an environmental control system; a cabin airflow delivery system in airflow communication with the environmental control system for receiving a cabin inlet airflow from the environmental control system; a cabin exhaust delivery system configured in airflow communication with the cabin of the aircraft for receiving a cabin exhaust airflow; and a fuel cell assembly in airflow communication with the cabin exhaust delivery system for receiving the cabin exhaust airflow.

(134) The aircraft of any preceding clause, wherein the fuel cell assembly is a polymer exchange membrane fuel cell assembly.

(135) The aircraft of any preceding clause, wherein the fuel cell assembly comprises a fuel cell having an anode and a cathode, and wherein the cathode of the fuel cell is in airflow communication with the cabin exhaust delivery system for receiving the cabin exhaust airflow.

(136) The aircraft of any preceding clause, wherein the environmental control system comprises a compressor, an air cycle machine, and an electric motor drivingly coupled to the compressor and the air cycle machine.

(137) The aircraft of any preceding clause, wherein the electric motor defines a maximum power draw, wherein the fuel cell assembly defines a maximum power rating, and wherein the maximum power rating is higher than the maximum power draw.

(138) The aircraft of any preceding clause, wherein the fuel cell assembly is in electrical

communication with the electric motor of the environmental control system for providing electrical power to the electric motor of the environmental control system.

(139) The aircraft of any preceding clause, wherein the environmental control system comprises a compressor and an air cycle machine, and wherein the environmental control system assembly further comprises a mixer, wherein the mixer is in airflow communication with the cabin airflow delivery system, wherein the environmental control system assembly further comprises a recirculation airflow path extending from the cabin to the mixer.

(140) A method of operating a fuel cell assembly comprising: providing a cabin exhaust airflow from a cabin of an aircraft through a cabin exhaust delivery system to a fuel cell assembly; and generating electrical power with the fuel cell assembly using the cabin exhaust airflow.

(141) The method of any preceding clause, wherein providing the cabin exhaust airflow from the cabin of the aircraft through the cabin exhaust delivery system to the fuel cell assembly comprises providing the cabin exhaust delivery system to the fuel cell assembly at a temperature of between 60 degrees Celsius and 90 degrees Celsius, and at a pressure of between 60 kilopascals and 100 kilopascals.

(142) The method of any preceding clause, further comprising: providing at least a portion of the electrical power generated to an environmental control system of the aircraft.

(143) The method of any preceding clause, further comprising: providing all of an electric power consumed by an environmental control system during a flight operation of the aircraft from the fuel cell assembly.

(144) An environmental control system assembly for an aircraft, the assembly comprising: an environmental control system; a fuel cell assembly in electrical communication with the environmental control system for providing electrical power to the environmental control system; and a controller operably connected to the fuel cell assembly, the controller operable to modulate an amount of power generated by the fuel cell assembly and provided to the environmental control system based on load forecasting data from an ECS load forecasting module of the controller.

(145) The assembly of any preceding clause, wherein the ECS load forecasting module determines the load forecasting data based on a passenger number for the aircraft, weather forecast data, historical data, or a combination thereof.

(146) The assembly of any preceding clause, wherein the ECS load forecasting module determines the load forecasting data based on a passenger number for the aircraft.

(147) The assembly of any preceding clause, wherein the ECS load forecasting module defines an anticipated forecast timing of at least five minutes ahead and up to one hour ahead.

(148) The assembly of any preceding clause, wherein the ECS load forecasting module comprises a load forecasting model.

(149) The assembly of any preceding clause, wherein the environmental control system is in airflow communication with a cabin of the aircraft, and wherein the fuel cell assembly is in airflow communication with the cabin of the aircraft for receiving a cabin exhaust airflow from the cabin.

(150) The assembly of any preceding clause, wherein the environmental control system comprises a compressor section, and wherein the fuel cell assembly is in airflow communication with the compressor section of the environmental control system for receiving a pressurized airflow from the environmental control system.

(151) The assembly of any preceding clause, wherein the controller further comprises an objective function module, and wherein the controller is operable to modulate the amount of power generated by the fuel cell assembly and provided to the environmental control system based at least in part on the objective function module.

(152) The assembly of any preceding clause, wherein the objective function module is operable between one or more objective functions, the one or more objective function selected from the following objective functions: maximum efficiency, maximum power output, maximum component life, and minimum emissions output.

(153) The assembly of any preceding clause, wherein to modulate the amount of power generated by the fuel cell assembly and provided to the environmental control system the controller is operable to modify an operating condition of the fuel cell.

(154) The assembly of any preceding clause, wherein the operating condition comprises a fuel cell temperature, a current output, a fuel utilization, an air utilization, or a combination thereof.

(155) The assembly of any preceding clause, wherein the controller is operable to modulate the amount of power generated by the fuel cell assembly and provided to the environmental control system based on a feedback control scheme and a feedforward control scheme.

(156) A method of operating an environmental control system assembly for an aircraft, the method comprising: determining ECS load forecasting data for an environmental control system of the environmental control system assembly; and controlling a fuel cell assembly in electrical communication with the environmental control system to modulate an amount of power generated by the fuel cell assembly and provided to the environmental control system in response to the determined ECS load forecasting data.

(157) The method of any preceding clause, wherein determining the ECS load forecasting data for the environmental control system comprises receiving data from a source external from the environmental control system assembly.

(158) The method of any preceding clause, wherein determining the ECS load forecasting data for the environmental control system comprises receiving data indicative of a passenger number for a flight operation of the aircraft.

(159) The method of any preceding clause, wherein determining the ECS load forecasting data for the environmental control system comprises determining the ECS load forecasting data for the environmental control system with an ECS load forecasting module.

(160) The method of any preceding clause, wherein determining the ECS load forecasting data for the environmental control system with the ECS load forecasting module comprises determining the ECS load forecasting data for the environmental control system with a load forecasting model of the ECS load forecasting module using load forecasting factors.

(161) The method of any preceding clause, wherein determining ECS load forecasting data for the environmental control system of the environmental control system assembly comprises determining the ECS load forecasting data based on a passenger number for the aircraft, weather forecast data, historical data, or a combination thereof.

(162) The method of any preceding clause, wherein determining ECS load forecasting data for the environmental control system of the environmental control system assembly comprises determining the ECS load forecasting data based on a passenger number for the aircraft.

(163) The method of any preceding clause, wherein the environmental control system is in airflow communication with a cabin of the aircraft, and wherein the fuel cell assembly is in airflow communication with the environmental control system, the cabin of the aircraft, or both.

(164) The assembly of any preceding clause, wherein the fuel cell assembly includes a fuel cell in direct airflow communication with a compressor of the environmental control system.

(165) This written description uses examples to disclose the present disclosure, including the best mode, and also to enable any person skilled in the art to practice the disclosure, including making and using any devices or systems and performing any incorporated methods. The patentable scope of the disclosure is defined by the claims, and may include other examples that occur to those skilled in the art. Such other examples are intended to be within the scope of the claims if they include structural elements that do not differ from the literal language of the claims, or if they include equivalent structural elements with insubstantial differences from the literal languages of the claims.

Claims

1. An environmental control system assembly for an aircraft, the assembly comprising: an environmental control system; a fuel cell assembly in electrical communication with the environmental control system for providing electrical power to the environmental control system; and a controller operably connected to the fuel cell assembly, the controller operable to modulate an amount of power generated by the fuel cell assembly and provided to the environmental control system based on load forecasting data from an ECS load forecasting module of the controller.
2. The assembly of claim 1, wherein the ECS load forecasting module determines the load forecasting data based on a passenger number for the aircraft, weather forecast data, historical data, or a combination thereof.
3. The assembly of claim 1, wherein the ECS load forecasting module determines the load forecasting data based on a passenger number for the aircraft.
4. The assembly of claim 1, wherein the ECS load forecasting module defines an anticipated forecast timing of at least five minutes ahead and up to one hour ahead.
5. The assembly of claim 1, wherein the ECS load forecasting module comprises a load forecasting model.
6. The assembly of claim 1, wherein the environmental control system is in airflow communication with a cabin of the aircraft, and wherein the fuel cell assembly is in airflow communication with the cabin of the aircraft for receiving a cabin exhaust airflow from the cabin.
7. The assembly of claim 1, wherein the environmental control system comprises a compressor section, and wherein the fuel cell assembly is in airflow communication with the compressor section of the environmental control system for receiving a pressurized airflow from the environmental control system.
8. The assembly of claim 1, wherein the controller further comprises an objective function module, and wherein the controller is operable to modulate the amount of power generated by the fuel cell assembly and provided to the environmental control system based at least in part on the objective function module.
9. The assembly of claim 8, wherein the objective function module is operable between one or more objective functions, the one or more objective functions selected from the following objective functions: maximum efficiency, maximum power output, maximum component life, and minimum emissions output.
10. The assembly of claim 1, wherein the controller is operable to modify an operating condition of the fuel cell assembly to modulate the amount of power generated by the fuel cell assembly and provided to the environmental control system.
11. The assembly of claim 10, wherein the operating condition comprises a fuel cell temperature, a current output, a fuel utilization, an air utilization, or a combination thereof.
12. The assembly of claim 1, wherein the controller is operable to modulate the amount of power generated by the fuel cell assembly and provided to the environmental control system based on a feedback control scheme and a feedforward control scheme.
13. A method of operating an environmental control system assembly for an aircraft, the method comprising: determining ECS load forecasting data for an environmental control system of the environmental control system assembly; and controlling a fuel cell assembly in electrical communication with the environmental control system to modulate an amount of power generated by the fuel cell assembly and provided to the environmental control system in response to the determined ECS load forecasting data.
14. The method of claim 13, wherein determining the ECS load forecasting data for the environmental control system comprises receiving data from a source external from the environmental control system assembly.
15. The method of claim 13, wherein determining the ECS load forecasting data for the environmental control system comprises receiving data indicative of a passenger number for a

flight operation of the aircraft.

16. The method of claim 13, wherein determining the ECS load forecasting data for the environmental control system comprises determining the ECS load forecasting data for the environmental control system with an ECS load forecasting module.

17. The method of claim 16, wherein determining the ECS load forecasting data for the environmental control system with the ECS load forecasting module comprises determining the ECS load forecasting data for the environmental control system with a load forecasting model of the ECS load forecasting module using load forecasting factors.

18. The method of claim 13, wherein determining ECS load forecasting data for the environmental control system of the environmental control system assembly comprises determining the ECS load forecasting data based on a passenger number for the aircraft, weather forecast data, historical data, or a combination thereof.

19. The method of claim 13, wherein determining ECS load forecasting data for the environmental control system of the environmental control system assembly comprises determining the ECS load forecasting data based on a passenger number for the aircraft.

20. The method of claim 13, wherein the environmental control system is in airflow communication with a cabin of the aircraft, and wherein the fuel cell assembly is in airflow communication with the environmental control system, the cabin of the aircraft, or both.
